

Discovery Executive Summary

Project: discovery- Dev Env · Generated: 7/16/2026, 11:53:22 AM

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating	Hotspot Score
1	Frontend Modernization Analysis	HIGH RISK	—

1. Frontend Modernization Analysis

OVERALL CODEBASE RATING — FRONTEND MODERNIZATION

High Risk

Driven by H8 (accessibility gaps — 57/57 social-media-react components lack ARIA attributes) with supporting Moderate ratings on duplication, component size, global state coupling, and prop drilling.

EXECUTIVE SUMMARY

The `target/` workspace contains two independent React 18 applications with divergent modernization postures. `workbench-demo` is a small TypeScript app using React Router v6, Tailwind CSS, and accessible form patterns; `social-media-react` is a larger JavaScript CRA app with 57 functional components, legacy Redux (`createStore` + thunks), React Router v5 (`Switch`, `useHistory`, `component=` routes), and widespread Redux coupling. No class-based components were found — adoption of function components with hooks is 100%. The primary risks are missing accessibility across the social app (0 `aria-*` attributes in 57 components), moderate UI duplication in user-card preview components (~7%), legacy global Redux patterns (47.5% of components read global store), and messaging prop-drilling depth of 3. Largest single component is `Message.jsx` at 250 LOC (moderate, below the 500 LOC threshold).

3.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	UI Component Duplication	Duplicate components %	<5%	5–10%	>10%	6.6% (4/61)	MODERATE
H2	Legacy Class-Based Components	Modern component adoption %	>90%	70–90%	<70%	100%	GOOD
H3	Massive Components	Largest component LOC	<200	200–500	>500	250 LOC	MODERATE
H4	Global State Dependencies	Components reading global state %	<30%	30–60%	>60%	47.5% (29/61)	MODERATE
H5	Complex State Management	Max prop-drilling depth	<3	3–5	>5	3 levels	MODERATE
H6	Direct DOM Manipulation (additional)	Components using imperative DOM APIs %	<5%	5–15%	>15%	3.3% (2/61)	GOOD
H7	Legacy Redux Store (additional)	Apps using plain <code>createStore</code> vs RTK %	0%	1–50%	>50%	50% (1/2 apps)	MODERATE
H8	Accessibility Gaps (additional)	Social-app components lacking any <code>aria-*</code> %	<20%	20–50%	>50%	100% (57/57)	HIGH RISK

3.5 Actions Required

Hotspot	Action	Rating	Priority
H8 Accessibility Gaps	Add <code>eslint-plugin-jsx-ally</code> to <code>social-media-react</code> ; refactor <code>Home.jsx</code> and <code>Signup.jsx</code> forms to match <code>workbench-demo/LoginPage.tsx</code> label/ARIA patterns; add <code>aria-label</code> to all icon-only buttons	HIGH RISK	CRITICAL
H1 UI Component Duplication	Extract <code>UserCard</code> and <code>useReactionToggle</code> shared modules; refactor 4+ preview components to compose them	MODERATE	HIGH
H4 Global State Dependencies	Migrate <code>social-media-react/src/store/</code> to Redux Toolkit slices; remove <code>window.myBus</code> ; add domain selector hooks	MODERATE	HIGH
H3 Massive Components	Extract <code>useChat</code> hook from <code>Message.jsx</code> ; split <code>CommentPreview.jsx</code> into subcomponents; enforce 200 LOC soft limit	MODERATE	MEDIUM
H5 Complex State Management	Introduce <code>MessagingContext</code> or extend <code>chatModule</code> slice to eliminate 8-prop drill through <code>Messaging</code> / <code>ListMsg</code>	MODERATE	MEDIUM
H7 Legacy Redux Store	Replace <code>createStore</code> + manual reducers with <code>configureStore</code> and <code>createSlice</code> across 4 modules	MODERATE	MEDIUM

3.6 Expected Outcomes

- A shared `UserCard` and form-field library reduces preview-component duplication from 6.6% toward the <5% Good band and standardizes loading/error UX.
- Redux Toolkit slices and selector hooks (`useLoggedInUser` , `useMessaging`) cut global-store coupling below 30% and make state transitions traceable.
- Extracting `useChat` and a `MessagingContext` eliminates 3-level prop drilling and enables isolated unit tests for chat flows.
- Adopting `workbench-demo` accessibility patterns across `social-media-react` closes the WCAG compliance gap on login, navigation, and interactive post/comment flows.
- Aligning both apps on React Router v6 and a shared auth composable enables a single design-system entry point for future HR feature modules.

Full report saved to `target/docs/discovery/03-frontend-modernization.md` (pipeline artifact copy at `agent-runs/20260716T114742_k6rmmk/03-frontend-modernization.md`). The orchestration UI will convert the target file to PDF automatically.